

Monitoring chloride diffusion in BFS concrete using an Electrical Resistivity Tomography device

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Abstract

Corrosion of steel bars in Reinforced Concrete (RC) due to the penetration of chlorides coming from seawater is one of the main pathologies leading to degradation of marine structures. Destructive Evaluation (DE) methods like coring from concrete in such marine environment are difficult and expensive for the survey of marine RC structures. Whereas, Non-Destructive Evaluation (NDE) methods based on the electric and electromagnetic wave propagation can reduce the number of cores necessary to evaluate the health of the studied structures. The aim of this work is to study the chloride ingress into saturated concrete by using the NDE methods, especially the Electrical Resistivity Tomography (ERT) device. Accordingly, chlorides diffusion campaign with two different sodium chloride concentration ($[NaCl] = 35$ and 90 g/L) was carried out on two slabs of concrete contains Blast Furnace Slag (BFS). Apparent resistivities obtained by ERT device were then inverted in order to obtain a resistivity profile versus depth with two different inversion software: Res1D and CERIS. Moreover, resistivity measurements were carried out regularly on the slabs to obtain the resistivity profiles. On the other hand, a calibration method was realized to relate the resistivity to the free chloride content. The free chloride profile obtained by NDE, after 12 weeks of diffusion, were then compared to that obtained by DE method.

Keywords: electrical resistivity, chloride diffusion, reinforcement corrosion, BFS concrete, Destructive Evaluation.

1 Introduction

The chlorides ingress is one of the essential factors induced to the deterioration of reinforced concrete structures in marine environment [1]. For the structures immersed permanently in seawater, chlorides penetrate into concrete through diffusion under a concentration gradient. Once the critical chloride concentration was reached at the reinforcement [2], the steel bars were depassivated leading to the beginning of corrosion [3]. By consequence, the corrosion mechanism leads to a reduction in the steel surface area, then a decrease in the mechanical resistance of the structure [4]. The duration preceding the initiation of corrosion is defined as incubation period according to Tuutti [3]. Thus, the assessment of the chloride profile into cover concrete during incubation period helps to evaluate the maintenance needs and to update the service life prediction of the structures. Indeed, the majority of realized studies to determine chloride content in concrete were carried out by the Destructive Evaluation (DE) method. This type of method known to be very exhausting, time-consuming and harmful for the structure. As well as, it is impossible to monitor the evolution of chloride content in the same position, because the concrete was destroyed by grinding.

The novelty of this work resides in using fast and reliable Non-Destructive Evaluation (NDE) method, to determine free chloride profile during a chloride diffusion campaign, especially the electric techniques. Several research studies had proved the sensitivity of this technique to the chloride content in concrete with Portland cement which presents low resistivity compared to BFS concrete [5–10]. In this paper, the Direct Current (DC) electrical resistivity method in Wenner configuration [7, 11–14] was used to study the efficiency of the electrical methods for



BFS concrete with very high electrical resistivity [15–20]. The Electrical Resistivity Tomography (ERT) device developed by Du Plooy et al. (2013) [21] was used in this work to measure the apparent electrical resistivity on BFS concrete. Consequently, the apparent resistivity measurements were submitted to an inversion model using two different software to estimate the “real resistivity” profile in concrete. Finally, the “real resistivity” profile obtained was converted to a free chloride concentration profile using an experimental calibration realized independently. The measurements were carried out on two slabs of BFS concrete submitted to a diffusion campaign with two different sodium chloride concentration (35 and 90 g/L) to obtain the free chloride profile at 11 different exposure periods. The free chloride profile obtained by NDE, after 19 weeks of diffusion, were then compared to that obtained by DE method.

2 Material and methodology

2.1 Material and slab conditions

The concrete slabs used were casted and demolded in 2013. The slabs were completely immersed in tap water after their casting until the start date of the diffusion campaign in September 2019. The binder of BFS concrete is composed of 84% of CEM III/A 52.5 L PMES CP1 NF and 16% of Calcareous filler addition (see **Table 1**). The cement CEM III/A, containing 62% of BFS and 38% of clinker, is particularly used for structures in a maritime environment [22].

Table 1: Concrete mix design of concrete

Components	Aggregate		Sand		CEM III/A	Calcareous filler	Sika Prise	Effective water
	(6/10)	(11/22)	(0/2)	(0/4)				
Mass (kg/m ³ of concrete)	320	760	430	430	260	50	1.9	177

The chloride diffusion campaign was carried out on 2 slabs of BFS concrete (BFS-35 and BFS-90). Each slab was submitted to 2 different sodium chloride concentration (35 and 90 g/L). The campaign was realized under saturated conditions to ensure pure chloride diffusion. The slabs were placed separately in closed boxes where the bottom face was immersed up to 1 cm, at initial test time t_0 , in regularly renewed solutions with different sodium chloride concentration. A small basin filled with tap water had been created by taping the border of the top face with aluminum foil to prevent slab from drying. Then the apparent resistivity measurements were performed, using the protocol described in §2.2.1, for an exposure period of 1 day, 3 days, 1, 4, 10, 12, 14, 19, 21, 27 and 46 weeks for all the slabs.

2.2 NDE Methodology

2.2.1 ERT device

The ERT device is composed of a multi-electrode resistivity probe (fourteen hollow metallic electrodes) aligned and equally spaced (2 cm). The electrodes are filled with wetted sponges in order to enhance the electrical current injection by reducing contact resistance (see **Figure 1**). The measurement sequence for the Wenner configuration was programmed on a commercial resistivity meter (Syscal Pro, Iris Instruments®). Once the current intensity and the potential drop were obtained, the electrical resistance R (Ω) can be determined using Ohm’s law ($R=U/I$). The electrical resistance R is multiplied by a geometric factor G_R (m) in order to obtain the apparent resistivity ρ_a ($\Omega.m$). The geometric factor G_R is computed by numerical modeling with

a finite element software (COMSOL Multiphysics®) and by taking into account the slab geometry and the electrodes configuration [21]. As Wenner configuration was applied to the fourteen electrodes, four different electrodes spacings (2, 4, 6 and 8 cm) were possible, then four investigation depths (or level) were obtained. In this work, three set of measurements were realized at each depth (see **Figure 1**). The average of the three sets of measurement were calculated for each level and thus four apparent resistivity were determined.

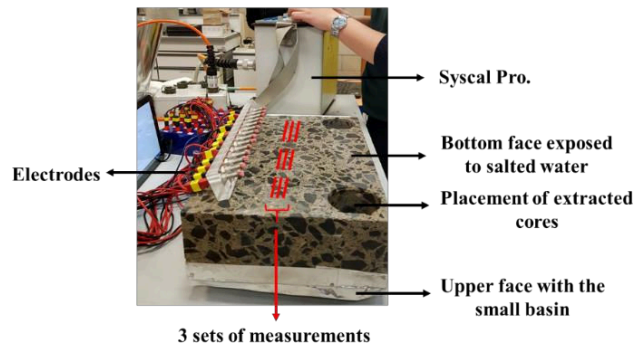


Figure 1: NDT-CE 2022

2.2.2 Apparent resistivity inversion

As mentioned previously, each apparent resistivity represents an integrated value over a volume related to electrodes spacing. Consequently, the apparent resistivities were inversed using two different methods in order to find the real resistivity profile versus depth. The first one was carried out using the free Res1D software [23]. Four resistivities values were determined by Res1D related to four layers with different thicknesses and then they were displayed at the center of each layer (5.1, 16.57, 30.92 and 44.45 mm). The second inversion method called CERIS developed to get moisture profiles [14,24] was carried out using an optimized model parameterization in order to increase the number of layers based on a realistic resistivity profile shape. The resistivity profile was adjusted with the Weibull curve equation.

2.2.3 Calibration : core conditioning

As mentioned before, a calibration campaign was realized to establish the calibration laws which are corresponding to the real resistivity versus the free chloride concentration. This calibration curve were used to convert the resistivity profile to the free chloride concentration. Six cores (H =70 mm ; D= 75 mm) were oven-dried at 60°C until mass stabilization and were then saturated under vacuum with solutions having 3 sodium chloride concentrations '[NaCl]=0, 35 and 90 g/L' (2 cores by concentration). After two days of cores saturation, the electrical resistivity measurements were realized using the resistivity cell [21]. Considering that the cores are homogeneous (same chloride content along depth), the measured electrical resistivity is considered as 'the real resistivity'. In order to determine the real free chloride concentration for each core which will represent the abscissa of the calibration curve, a DE campaign was carried out at the end of the resistivity measurements. The cores were sawn into three parts where each part was grounded with a vibrating disc mill RS200® to obtain a powder with 20 µm of fineness. The procedure recommended by RILEM association [25] was used to extract free chloride. Finally, the free chloride concentration was determined by potentiometric titration. The average of the 3 free chloride concentrations was used to determine the abscissa of calibration curves.

2.2.4 Validation of chloride profile obtained by NDE

In order to validate chloride profiles obtained by NDE, a DE campaign was carried out to find the ‘real free chloride profile’ considered as reference profile. One core was extracted from each slab after 12 weeks of exposure. These cores were grounded in several steps of an average of 5 mm for each step, perpendicular to the bottom face exposed to NaCl solution, using a grinding instrument (German instrument®) as recommended by Vennesland et al. (2013) [26]. Then, the free chloride concentration to calibration cores was determined following the same procedure used to calibration cores.

3 Results

3.1 Calibration results

As expected, the electrical resistivity decreases progressively for BFS concrete with the increase of the chloride concentration (c.f. **Figure 2**). The resistivity measurement displays a slow decrease for high chloride concentration ($[\text{NaCl}] = 90 \text{ g/L}$) while a fast decrease for low chloride concentration ($[\text{NaCl}] = 35 \text{ g/L}$) for both concrete, similarly to the results of Hunkeler (1996), Saleem et al. (1996) and Fares et al. (2018) [5,8,27,28]. The calibration curves were obtained by fitting the experimental resistivity values using a power function [29] with a regression coefficient superior to 0.9.

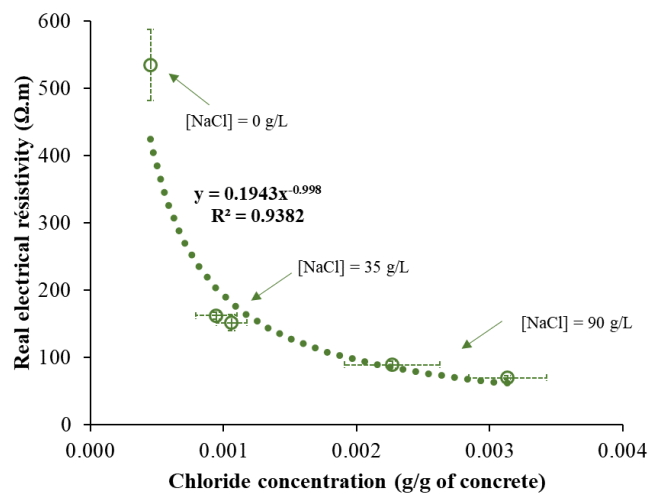


Figure 2: Calibration curves of electrical resistivity as a function of free chloride concentration

3.2 The apparent resistivity results

The apparent resistivity values were obtained for the two slabs (BFS-35 and BFS-90) during the diffusion campaign. For example, the apparent resistivities for slab BFS-35 versus diffusion time are presented in **Figure 3**. As the slabs were saturated under vacuum with tap water to obtain a homogeneous slab (same chloride and water content along depth), thus the apparent electrical resistivities measured by the ERT device at initial test time t_0 for the 4 electrodes spacings (or in other words for the 4 depths) were expected to be equal. However, the apparent electrical resistivity at level 1 (nearest Level to the external ambient) is slightly higher than those of other deeper levels (2, 3 and 4) which indicates an initial property gradient in the concrete. For both slab, the apparent resistivity values for the 4 levels decreased instantly, as

expected, with the beginning of the diffusion campaign due to the penetration of chlorides ions. Moreover, within a few days of chloride diffusion, the highest apparent resistivity is located at the nearest Level to the external surface (Level 1 or $a=2\text{cm}$) whereas the lowest resistivity is located at the farthest Level to the external surface (Level 4 or $a=8\text{cm}$). The increase of the resistivity in-depth reveals a higher chlorides concentration on the surface and a lower concentration in the depth which agrees with the direction of chlorides ions propagation. However, between the first and the sixth month of diffusion, an increase of the apparent resistivity is noticed for the 3 Levels (2, 3 and 4) due to the late binding of chloride with the cement matrix and the formation of Friedel's salt.

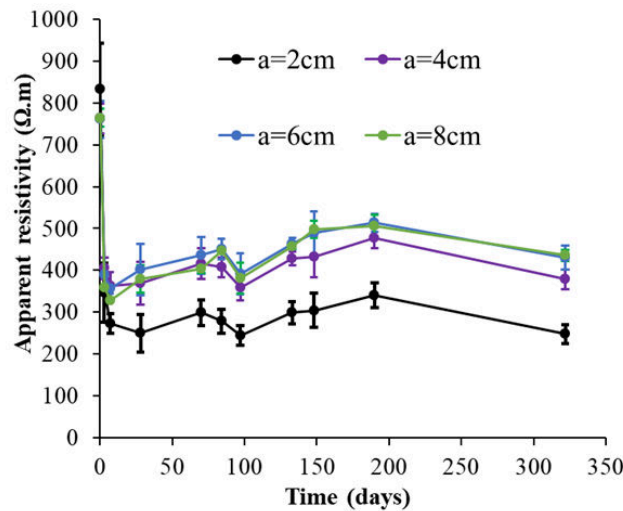


Figure 3: Apparent resistivity for slab BFS-35

3.3 Real resistivity profiles

The real resistivity profiles for the slab BFS-35 obtained by Res1d and CERIS inversion are presented in **Figure 4.a** and **Figure 4.b**, respectively. As already mentioned, the resistivity profiles inverted by Res1D are finally represented in 4 layers of different thicknesses while those inverted by CERIS are represented in the form of curves thanks to the continuous parameterization.

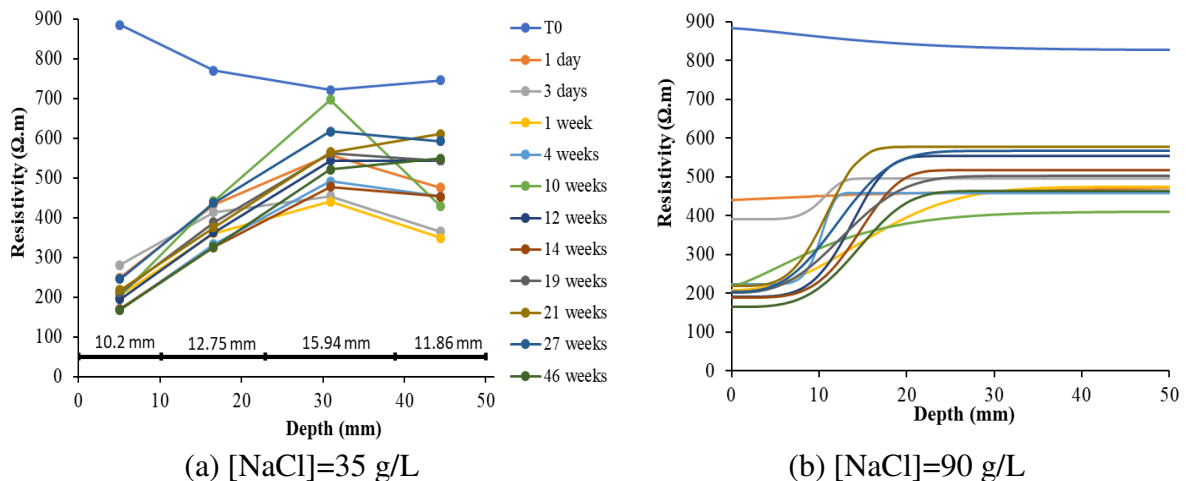


Figure 4: Real resistivity profile for slab BFS-35 obtained: (a) inversion with Res1D, (b) inversion with CERIS

The profiles obtained by the CERIS or Res1D show a quite logical evolution where we can clearly observe the decrease in the surface resistivity over time indicating an increasing in the chloride content. The resistivity profiles obtained for the other slab have the same behavior during the diffusion campaign but they will not be presented in this paper. These profiles will be converted in the next section to free chlorides profiles by using the calibration curves displayed in **Figure 3**.

3.4 Free chloride profiles

The free chloride profiles obtained by NDE and DE, after 12 weeks of diffusion, for both slab BFS-35 and BFS-90 are presented in **Figure 5.a** and **Figure 5.b**, respectively. Each figure presents three chloride concentration profiles: (1) a profile obtained by NDE with CERIS inversion presented with a smooth solid line, (2) a second profile obtained by NDE with Re1D displayed with straight and marked solid line, (3) and a third profile obtained by DE presented with a straight and marked dash line. For a higher exposure concentration of NaCl equal to 90 g/L, it could be observed that the chloride profiles at the surface (0 to 10 mm) is higher than that obtained for the other slab BFS-35. The high concentration of chlorides in the external solution increases the contrast in chloride concentration gradient which induces to accelerate the diffusion process.

Regarding the comparison between the NDE chloride profiles obtained by two inversion method (Res1D-CERIS), the results show a very good similarity between the two profiles with very small deviation. This small deviation due to the discontinuity of the Res1D profiles is more visible when passing from one layer to another. Moreover, the NDE chloride profiles (CERIS or Res1D) show a good compatibility with those obtained by DE, especially for the slabs submitted to high chloride concentration (NaCl=90 g/L) where chloride gradient is more contrasted with an ERT device more sensible.

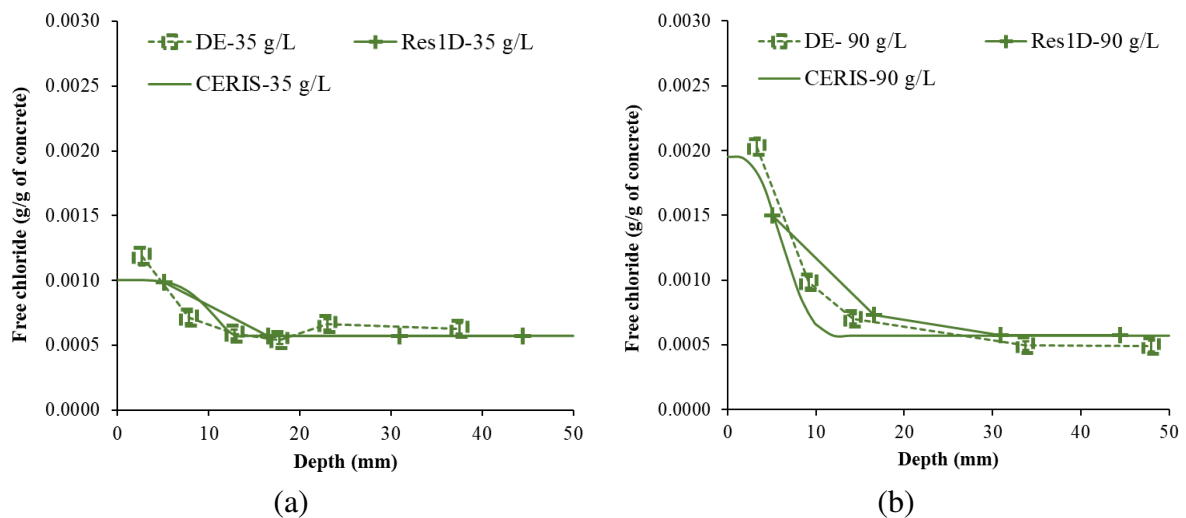


Figure 5: Free and total chloride concentration profile obtained by DE and NDE (Res1D and CERIS) after 12 weeks of diffusion for slab: (a) BFS-35, (b) BFS-90

Moreover, the RMSE 'Root-Mean-Square Error' was calculated to quantify the difference between the DE profiles considered as 'reference profiles' and the NDE profiles. Indeed, having a lower value of RMSE means that the similarity between DE and NDE profile is higher and the ND inversion errors are less important. The RMSE values are in very good agreement with

the difference between the DE and NDE profiles displayed in **Figure 6**, we can notice that the RMSE values for the profiles obtained using CERIS program are lower than those obtained using Res1D. The reducing in the RMSE value reveals the good improvement in the chloride profiles obtained by NDE thanks to CERIS program which the NDE profiles become closer to the ‘reference profile’ or more realistic.

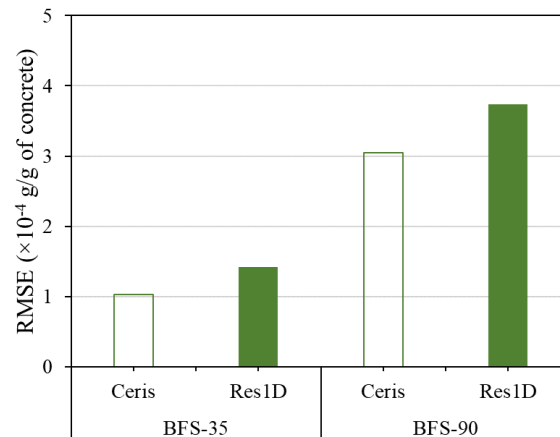


Figure 6 : RMSE for free chlorides profiles obtained by DE and NDE (CERIS and Res1D)

4 Conclusion

In this work, the efficiency of ERT device to determine free chloride profile was studied for BFS concrete slabs, having a high electrical resistivity, during a diffusion campaign. The ND methodology entails converting the electrical resistivity profiles obtained by the surface electrical resistivity measurements and two different inversion processes to a free chloride content profile by using calibration curves found independently. The comparison of the chloride profiles obtained by DE and NDE displays a good similarity even for both slab with a higher compatibility with that submitted to a higher chloride concentration (BFS-90). Among the two-inversion method used in this work, a new one called CERIS was applied in order to improve the reliability of the results by having a continuous resistivity profile. The RMSE values reveal the positive effect of using CERIS inversion program by making the chloride profiles obtained by NDE closer to the ‘reference profile’.

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